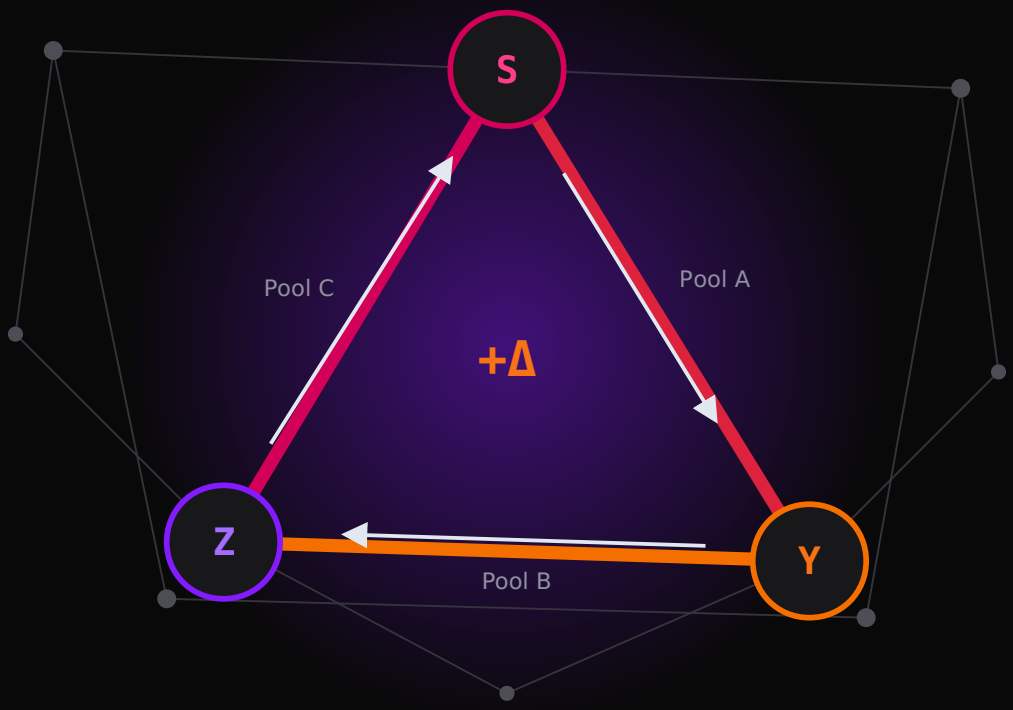
PULSECHAIN · ON-CHAIN ARBITRAGE SCANNER

Arb-Me-Now

Process Manual & White Paper

Methodology, operating procedure, verified parameters and risk model



PulseX v1 · v2 | 9mm v2 | LibertySwap V3 | 2-pool + triangular | BigInt engine | read-only

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cryptales.github.io/Arb-Me-Now

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At a glance. Arb-Me-Now reads every pool from one PulseChain block, models each swap with exact integer maths, finds the optimal size for 2-pool and triangular routes, and reports profit before and after estimated gas. It never connects a wallet or sends a transaction.

1. Purpose

Arb-Me-Now finds price differences between decentralised exchanges (DEXs) on PulseChain and tells you whether a round-trip trade would make money **after DEX fees**. It reads live pool data straight from the chain, so all numbers come from a single block. It never signs or sends transactions, and it never asks for a wallet or private key.

2. Architecture

Layer	What it does
Data	Public RPC `rpc.pulsechain.com`. All pool states are read in **one** Multicall3 `tryBlockAndAggregate` call (selector `0x399542e9`), so every price comes from the same block. PulseChain RPC rejects JSON-RPC batch requests, so Multicall3 is required.
Tick data (v4.5, adaptive since v4.8)	A second Multicall3 call, pinned to the **same block**, reads `ticks(int24)` (selector `0xf30dba93`) for an adaptive number of boundary ticks on each side of every v3 pool (see §3.9; cap 1–8, default 4), giving `liquidityNet` at each crossing. If this read fails, the engine falls back to single-interval maths.
Discovery	`getPair` / `getPool` lookups on each factory for a fixed list of liquid pairs over the watch list (v4.7: WPLS, PLSX, HEX, eHEX, INC, WETH, DAI and bridged stables; new pairs WPLS/eHEX, HEX/eHEX, eHEX/DAI, WPLS/INC, PLSX/INC, INC/HEX, WETH/DAI, WETH/PLSX, WETH/HEX).
Engine	Pure BigInt maths (`engine.js`) with exact integer swap formulas and no floating-point rounding in the money path. It is unit-tested in Node (`test.js`).
UI	A single self-contained `index.html` hosted on GitHub Pages, with no backend.
External	LibertySwap quote API (bridge checks) and Coinbase spot (ETH/USD reference only).

Venues and verified fees

Venue	Factory	Fee
PulseX v1	`0x1715a3E4A142d8b698131108995174F37aEBA10D`	0.29%
PulseX v2	`0x29eA7545DEf87022BAdc76323F373EA1e707C523`	0.29%
9mm v2	`0x3a0Fa7884dD93f3cd234bBE2A0958Ef04b05E13b`	0.25%
LibertySwap V3	`0x796fcbDC956b85797EFe21145Aa97599B7FB36a6`	0.01 / 0.05 / 0.25 / 1% tiers

All fees were checked on-chain, not taken from documentation.

3. Methodology

3.1 Swap maths

- **v2 (constant product):** `out = in·(10000−fee)·R_out / (R_in·10000 + in·(10000−fee))`, using integer division the same way the pool contract does.
- **v3 (concentrated liquidity, multi-tick since v4.5):** exact swap maths using `sqrtPriceX96` and active liquidity `L`, walking across up to **N tickSpacing intervals per direction** (N adaptive since v4.8, §3.9; cap 1–8, default 4). When the price crosses a boundary tick, `liquidityNet` is added (moving up) or subtracted (moving down) to get the next interval's `L`. The fee is taken on the gross input. Any trade that would go beyond the last interval read, or into an interval with zero liquidity, is treated as infeasible. If tick data is unavailable, the engine falls back to the conservative single-interval model.

3.2 Safe-depth caps

- A v2 leg may use at most **10% of the input reserve**.
- A v3 leg may not move the price beyond the **4th tick interval** read in its direction (or the current interval in single-interval fallback).

3.3 Route types

1. **Two-pool cycles:** S → Y on pool A, then Y → S on pool B, where A and B are different pools for the same pair.
2. **Triangular 3-hop routes (v4.4):** S → Y → Z → S across three different pools.

3.4 Optimal sizing

For each route, the engine first finds the largest feasible input size by binary search. It then runs a ternary search for the size with the highest profit, which works because profit rises and then falls as size grows. Finally it checks every size in the last narrow window exhaustively. It also evaluates the size you typed in, so you can compare your size with the best one.

3.5 Filtering and pricing

- **Suspicious/dust:** routes where any leg has under the dust threshold in USD depth, or an implausibly large price gap, are hidden by default.
- **USD values:** WPLS/USD comes from the deepest WPLS/stablecoin pool, treating bridged stables as $1. Other tokens are priced through their deepest WPLS pool.
- **Snapshot age:** data is marked stale after 20 seconds.

3.6 Bridge price check (LibertySwap)

`GET apis.libertyswap.finance/v3/swap/quote` gives the real quote from PulseChain to Ethereum (fee 0.5% for USDC and WETH from 369 → 1). The result is compared with the PulseChain purchase price and the Coinbase spot price. **This is not atomic:** the bridge takes minutes and you carry price risk while it runs. The API rate limit is 30 requests per 60 seconds and is enforced on the client.

3.7 Atomic execution: ArbExecutor (v4.6)

`contracts/src/ArbExecutor.sol` (Solidity 0.8.24, Foundry) runs a 2- to 4-leg closed cycle S → … → S in **one transaction**: - **Flash swap on leg 0:** the first pool sends its output first; the remaining legs run inside its callback; leg 0 is repaid in S at the end. No starting capital is needed. - **All-or-nothing:** if the cycle returns less than the amount owed it reverts `NoProfit(returned, required)`; it also reverts if the leftover is below `minProfit`. Only gas can be lost. Profit is sent to the owner. - **Callbacks (verified in on-chain bytecode):** PulseX v1/v2 `uniswapV2Call`, 9mm v2 `pancakeCall`, LibertySwap V3 `libertyV3SwapCallback` (`0x31e2ba55`). - **Safety:** `execute` by the owner or the optional keeper (v4.9, §3.11); `withdraw` / `setKeeper` owner-only; callbacks accepted only from the pool currently being swapped while a run is active; deadline; token-continuity check; fee-on-transfer tokens unsupported. - **Leg encoding (uint256):** bits 0–159 pool · 160–167 kind (0 v2, 1 v3) · 168–175 zeroForOne · 176–191 v2 fee bps. Call: `execute(uint256[] legs, uint256 amountIn, uint256 minProfit, uint256 deadline)` (selector `0x29df2c6f`). - **Testing:** 11/11 Foundry tests on a PulseChain fork (live route reverts NoProfit; induced gaps profitable for v2→v2, v3-flash→v2, v2→v3 and a 3-hop; access control, stray callbacks, expiry, open cycles). A JS end-to-end test on an anvil fork drove the scanner's own encoder: contract profit matched the engine's prediction **exactly** for v2→v2 and within **2 wei** with a v3 leg, and a real execute tx paid the owner. Measured `execute` gas (Foundry report): median ≈ 294k, max ≈ 612k.

3.8 Wallet send (v4.7)

Trade Companion's opt-in **Connect & Send** uses the injected EIP-1193 wallet (MetaMask): `eth_requestAccounts` → `wallet_switchEthereumChain` to `0x171` (369) → `owner()` read on the executor; sending is blocked unless the connected account is the owner → `eth_sendTransaction` with the built `execute(...)` calldata → the tx hash is shown with an explorer link. Keys never leave the wallet.

3.9 Adaptive tick depth (v4.8)

For each V3 pool the engine first computes the current interval's capacity *C* in start-token (S) units: the smaller of `v3MaxIn` with S as input and `v3MaxIn` with the other token as input, converted to S at the pool's sqrtPrice. It then reads `d = clamp(⌊size / C⌋ + 2, 2, cap)` tick intervals per direction, where *cap* is the user's slider value (1–8; cap 1 means single-interval maths). Pools that do not contain S (middle legs of 3-hop routes) read the full cap. Live check (2026-11-01, WPLS start): at 1,000 WPLS deep pools read 2 intervals; at 10^8 WPLS they read 6–8; with cap 1 every pool reads 1.

3.10 Piteas cross-check and the Tide venue (v4.8.1)

Piteas (piteas.io) is a DEX *aggregator*, not a DEX: it holds no liquidity and routes swaps across PulseChain venues through its router `0x6BF2…88A6`. Trade Companion's opt-in **Check with Piteas** button asks `GET https://sdk.piteas.io/quote` for the cycle's first hop (S → first intermediate token, at your input size) and then the way back (intermediate → S), i.e. 2 requests. It shows both outputs, the venues used, and the Piteas round trip against your input and against our route's output: - **Piteas round trip ≥ our route:** the gap is likely available market-wide, so other traders can take it too. - **Our route beats it:** the gap looks specific to the pools we found.

Piteas allows **10 requests per minute** (going over can block the IP for an hour) and states the API is not meant for arbitrage bots. So the check is manual only, capped locally at 6 requests per rolling 60 s, and never part of the automatic scan. The scanner does not depend on Piteas; if the API is down (it

returned HTTP 500 for a period on 3 Nov 2026), the panel shows "Piteas unavailable" and everything else keeps working. Piteas quotes cannot be an ArbExecutor leg: they have no flash callback, and their calldata is built off-chain for one exact amount.

Tide appeared in a Piteas route. On-chain inspection (3 Nov 2026) shows the pool `0x6926…E38B` ("RISKOFF") is a **Balancer V3-style weighted pool**, not a Uniswap-style pair: it exposes `getVault()` (vault `0x634f…4b2d`), `onSwap`, `getWeightedPoolImmutableData` and `getCurrentLiveBalances`, and reverts on `getReserves()`. It holds 5 tokens (HEX, INC, CST, PLSX, WPLS) at weights 5/5/80/5/5 %, with a 0.30 % swap fee. Its live balances were small (≈14.8k HEX, 74 INC, 61k CST, 4.49M PLSX, 3.26M WPLS). Adding it would need weighted-pool maths in the engine **and** a Balancer-V3 vault integration in ArbExecutor, for very little depth, so it is **not scanned** for now. The Piteas panel labels Tide hops "(not scanned here)".

3.11 Deploy kit and keeper role (v4.9)

Keeper role. The `owner` is fixed at deployment (immutable, no transfer). Since v4.9 the owner can appoint one optional **keeper**, a separate bot key, with `setKeeper(address)`; `setKeeper(0x0)` revokes it. - The keeper may call `execute()` only. `withdraw` and `setKeeper` revert `NotOwner` for it. - Profit is always transferred to the **owner**, never to the caller. - Unauthorised callers get `NotAuthorized` (`0xea8e4eb5`); the zero address can never act as keeper. - So a leaked keeper key can at worst waste its own gas on reverting trades. It cannot move funds. - Fork tests (14/14 pass, 3 new): keeper executes a profitable cycle and the owner is paid while the keeper's balance is unchanged; keeper cannot withdraw or set keeper; a revoked keeper is rejected.

Deploy kit. `foundry.toml` pins solc 0.8.24, EVM `paris` (no newer opcodes, for maximum PulseChain compatibility), optimizer 200. The build embeds that exact creation bytecode (5,753 bytes, keccak `0x1635…1004`) in the page. In Trade Companion, after **Connect Wallet**: 1. **Estimate deploy cost:** `eth_estimateGas` of the creation tx from your address plus `eth_gasPrice`; nothing is sent. On 3 Nov 2026: about 1.56M gas at ~320k gwei, roughly 500 PLS. 2. **Deploy ArbExecutor:** a confirm dialog warns that the connected account becomes the permanent owner. The wallet signs the creation tx (gas = estimate + 20%). The page waits for the receipt, reads `owner()` back, and saves the address in this browser. 3. **Set / revoke keeper:** owner-checked before sending; the owner's own address is refused as keeper.

End-to-end fork test (3 Nov 2026, local anvil fork of PulseChain at block ~27.97M, injected test wallet; nothing touched mainnet). - Estimate: 1,297,519 gas, about 367 PLS. - Deploy from the owner address succeeded, and `owner()` matched. - Read keeper showed none, then Set keeper worked, and Dry run said "Would succeed, +52.86 WPLS". - The **keeper** sent the 3-leg route (7,122 WPLS flash input). Status 1: the owner received **52.86 WPLS** and the keeper received 0. - A stranger's Send and Set keeper were blocked by the page; Revoke worked. - **Important:** that execute used **770,688 gas**, about 218 PLS at 282,679 gwei. That is far more than the 52.86 WPLS profit (WPLS = PLS 1:1), so live it would have **lost about 165 PLS net**. It is also about 2.2× the scanner's 350k 3-hop gas assumption, so "net after gas" for routes with V3 legs was badly optimistic. **Fixed in v4.9.1** (§3.12).

Connect & Send now also accepts the keeper account. **Dry run** still simulates from the owner. The CLI alternative is `forge create … --interactive` on your own machine.

3.12 Gas model recalibration (v4.9.1)

The old flat estimates (250,000 gas for 2 hops, 350,000 for 3) came from generic DEX swaps. They did not include the ArbExecutor's flash-swap callbacks or the extra cost of V3 swaps. v4.9.1 replaces them with a per-route model based on measured gas:

Measured on a PulseChain fork (3 Nov 2026). The forge rows are the `execute` call only, with pools already warm from test setup, so they understate a real transaction.

Route (legs)	Measured gas
V2 flash + V2	383,256
V3 flash + V2	428,225
V2 flash + V3 (large size, tick crossings)	573,416
V2 + V2 + V2 triangular	346,153
3-leg route with V3 legs (full transaction, cold)	770,688

Model (v4.9.3): estimated gas = **210,000 base** (transaction + flash/callback overhead) + **140,000 per V2 hop** + **230,000 per V3 hop**. Values are rounded up to allow for cold storage access. v4.9.1 used a 150,000 base. It was raised by 60,000 after the first mainnet dry run measured 627,914 gas for a V2 × 3 route the old model put at 570,000 (§3.13). The new model gives 630,000.

Route	Pre-v4.9.1 (flat)	v4.9.1 (150k base)	v4.9.3 (210k base)
V2 + V2	250,000	430,000	490,000
V3 + V2	250,000	520,000	580,000
V2 × 3	350,000	570,000	630,000 (mainnet measured 627,914)
V3 + V3 + V2	350,000	750,000	810,000 (fork measured 770,688)
V3 × 3	350,000	840,000	900,000

- Each row's "net after gas" uses its own estimate and shows it, e.g. "(~580k gas)". The summary line states the model.
- **Dry run now measures real gas.** After a successful `eth_call`, the page also calls `eth_estimateGas` for the exact transaction from the owner and shows its gas, its cost in PLS/USD at the live gas price and, for WPLS routes, **net profit after gas**. That figure overrides the model.
- The model is still an estimate. Tick crossings, token transfer logic and storage warmth change real gas by tens of thousands. Always rely on the dry-run number before sending.

3.13 First mainnet deploy and live test (v4.9.2)

Deploy (verified on-chain, 4 Nov 2026). ArbExecutor was deployed from the owner's ZKX wallet with the in-app Deploy button at `0xd5da79af4c4c24f9d3afec0a1cbd761a24b130bd`. It used 1,297,519 gas. Checks run against https://rpc.pulsechain.com: `owner()` returns the deploying wallet, `keeper()` is

unset (0), and the creation bytecode's keccak matches the pinned build artifact. The runtime bytecode differs from the artifact only at the six offsets where the immutable owner is embedded, which is expected.

Dry run. A 3-leg V2 route (9mm 25 bps → 9mm 25 bps → PulseX 29 bps, 130.541 WPLS in, min profit 5.70 WPLS) returned "Would succeed" with **+11.396 WPLS before gas**. `eth_estimateGas` measured **627,914 gas**, about 233 PLS at 371,101 gwei, so the trade was **about −221.6 WPLS net**. The §3.12 model estimated 570,000 gas for this route type, about 9% low. **Fixed in v4.9.3:** the base was raised to 210,000, so this route is now estimated at 630,000. Dry-run gas is still the number to trust.

Live test tx `0x5e4ab9f20cd6274093e4f8ad8b3118c397d79473648d54088bfb885b8993d66d` (block 27,969,496). It was sent from the owner wallet to the executor and **reverted** with revert data `0x203d82d8` = `Expired()`. The calldata deadline was 1,793,771,000; the block timestamp was 1,793,771,995, so the tx landed **995 s (~16.6 min) after its deadline**. It used 23,723 gas at 410,031 gwei, so it **cost 9.73 PLS**. The wallet had shown a 301 PLS *maximum* fee; that is a cap, not the charge. No flash loan or swap ran.

What this proves: **ZKX can sign and broadcast `execute()` to the contract, and the owner check passes** (otherwise the revert would have been `NotAuthorized`). What failed was timing. The 120 s deadline was too short for the delay between clicking Send and the transaction being mined. The likely cause is the time spent in the wallet pop-up or in the wallet's queue; this is a hypothesis, not confirmed. Because the route was net-negative after gas, the revert actually avoided a loss of about 220 PLS.

Change in v4.9.2: the calldata deadline is now **now + 600 s**. The deadline is a safety net against stale prices, and `minProfit` still protects every trade, so the longer window doesn't add price risk to a trade that still meets min profit. Procedure: dry-run, click Send, and approve the wallet pop-up straight away. If it sat for more than a few minutes, reject it and rebuild.

3.14 Server scanner and Telegram alerts (v5.0)

Up to v4.9.4 the scanner only ran while the page was open in a browser tab. v5.0 adds `server/scanner.js`, a small Node.js program (Node 18 or newer, no npm packages) that runs the *same* engine (`engine.js`, the same file the web page uses) on an always-on machine. It sends a Telegram message whenever a route looks profitable after gas. This release only sends alerts. **It never signs or sends a transaction.** You still trade manually through the Trade Companion, which keeps the safety model unchanged.

What one scan does. It discovers pools on all four venues (PulseX V1, PulseX V2, 9mm V2 and LibertySwap V3), caches the list and refreshes it about every 120 scans. Then it reads every pool's live state in one Multicall3 round trip, fetches `eth_gasPrice`, prices WPLS and the start token in USD, and runs `scanCycles` (2-pool cycles) plus `scanTriangular` (3-hop cycles). Each row is filtered exactly as in the browser: - suspicious rows are dropped: any leg with under $25 of safe depth, or a mid gap over 50%; - rows below `min_profit_bps` are dropped (gross bps at the optimal size); - only rows whose **net USD after estimated gas is above zero** remain.

Scan log (v5.0.1). Alerts only fire on net-positive routes, but the interesting learning is in the near-misses. So from v5.0.1 the scanner appends one row per scan to `server/scan_log.csv` recording the single *best* route that scan — the executable route with the largest mid-price gap — **even when it is net-negative after gas**. "Executable" excludes dust mirages: non-suspicious 2-pool rows (already past the $25 safe-depth floor) plus triangular rows whose optimal size is at least $25. Each row carries the timestamp, scan and block number, gas price, how many routes were evaluated, the route

and its pools, the mid gap, the optimal size and its gross/net USD and bps, the gas estimate, and the profit at your fixed trade size. The file is git-ignored and controlled by `csv_log` / `csv_path` in `config.json` (on by default). Over days this builds a dataset of how close the market comes to a profitable cycle, which venues and pairs dominate the near-misses, and what gas levels would tip them positive — the raw material for tuning sizing, venue selection and the eventual keeper.

Gas uses the v4.9.3 calibration: 210k base + 140k per V2 hop + 230k per V3 hop, priced at the live gas price.

De-duplication. The same route (venue legs plus tokens) is alerted at most once every `dedup_minutes` (default 5), so a gap that lasts several blocks doesn't spam you.

The alert message shows the route, the pools for each leg, the mid gap, the optimal size, gross profit (in tokens, bps and USD), estimated gas and the net after gas, plus a link to the live site. The procedure is unchanged from §4: open the site, scan, check the row, Dry Run, then Send only if the dry run passes.

Configuration lives in `server/config.json`, which is git-ignored, so your bot token never gets committed. You create it by copying `config.example.json`. Fields: `telegram_bot_token` (from @BotFather), `telegram_chat_id`, `start_token`, `amount_in`, `min_profit_bps`, `scan_interval_s` (default 30), `dedup_minutes`, `v3_depth`, `do_triangular` and `rpc_urls`.

Commands. `node server/scanner.js` runs continuously. `node server/scanner.js --dry --once` runs a single scan and prints any alerts in the terminal instead of sending them; it needs no Telegram setup. Use it to check the install first. The full setup steps are in `server/README.md`.

Where to run it. It has to run somewhere that stays on: your own PC (in `screen` / `tmux`), a small VPS, or under `pm2`. A browser tab or a sleeping laptop will miss gaps.

Tested so far (2026-11-04): one live dry-run scan at block 27,970,424 evaluated 200 routes for 1,000 WPLS and found 0 net-positive, which matches the browser results that day. A real Telegram delivery has **not** been tested yet, because that needs your own bot token. Do the first live check with a test message after setup.

4. Operating Procedure (step by step)

1. Open https://cryptales.github.io/Arb-Me-Now/.
2. Under **Settings**, choose the start token and the trade size, and turn venues on or off.
   - **Min profit (bps of input):** the smallest return you care about, in basis points of the amount you put in (1 bp = 0.01%; 10 bps = 0.1%; 100 bps = 1%). It is checked against the profit at the **optimal size**, before gas. It only takes effect when **Only rows ≥ min profit** is ticked; otherwise every row is shown. The default is 0, so any profitable route passes. Example: 1,000 WPLS in, 1,002 WPLS out = 2 WPLS profit = 20 bps. The *LibertySwap bridge quotes* and *Triangular 3-hop routes* toggles are on by default.
3. Click **Scan**. Check the telemetry bar: RPC is **OK**, the snapshot block is shown, and the snapshot age is under 20 s.
4. Read the results table, which is sorted by profit in USD. Each row shows the route, venues, mid-price gap, best size, profit at the best size, profit at your size, and flags.
5. Click **Inspect** to open the Trade Companion. It shows reserves or v3 state for every leg, the depth caps, and the exact BigInt calculation.

6. **Before acting on any row:**
   - Re-scan to confirm the opportunity is still there in a fresh block.
   - Check the **net after gas** line. Gas is estimated from the live `eth_gasPrice` × a per-route estimate (210k base + 140k per V2 hop + 230k per V3 hop, §3.12), valued at WPLS/USD. The Trade Companion **Dry run** shows the measured `eth_estimateGas` figure and net after gas. Use that number before sending.
   - Allow for MEV and front-running risk. A public mempool trade can be sandwiched.
   - Check the pool addresses on scan.pulsechain.com.
7. **Alerts (v4.5):** tick 🔔 **Alert on new profitable route** and allow browser notifications when asked. Pair it with **Auto-refresh every 15 s**. The scanner beeps, shows a banner and sends a notification **once** for each newly profitable route that is ≥ your min bps and still positive after gas (suspicious rows never alert). Alerts only work while the tab stays open.
8. **Execute (v4.6, optional):** deploy your own ArbExecutor once, either with 🚀 **Deploy ArbExecutor** in Trade Companion (v4.9; signed in your wallet) or with `forge create src/ArbExecutor.sol:ArbExecutor --rpc-url https://rpc.pulsechain.com --interactive --broadcast` on your own machine. Optionally set a keeper (§3.11). In Trade Companion, enter its address, keep or edit size and min profit (default 50% of expected), click **Build calldata**, then **Dry run** (an `eth_call` from the owner; nothing is signed). If it says "Would succeed", send the calldata from your owner wallet (value 0) before the 600 s deadline (v4.9.2; was 120 s). Approve the wallet pop-up promptly: the deadline is fixed when you click Send, and a tx that lands later reverts with `Expired()`.
9. **Piteas cross-check (v4.8.1, optional):** in Trade Companion, click **Check with Piteas** to compare the route with the aggregator's whole-market round trip (2 requests; max 6 per minute). Use it sparingly.
10. For bridge checks, treat the edge as an upper bound. Gas on both chains, bridge time and slippage reduce it.

5. Interpreting results

5.1 Reading the results table (v4.9.4)

Each row is one route: a 2-pool cycle (S → Y → S) or a triangular 3-hop route (S → Y → Z → S). The first three columns are **#** (rank), **Cycle** (the token path) and **Buy leg → Sell leg** (the pools, in order). The other five columns work like this.

Mid gap. This is the raw price difference around the cycle, read from each pool's current state (V2 reserves, V3 `sqrtPriceX96`). It is calculated as the product of the leg mid-prices minus 1. It is **before DEX fees and before any price impact**, so it is the ceiling on what the route could ever earn. If the mid gap is smaller than the combined fees of the legs (for example, two PulseX hops = 0.58%), the route can't be profitable at any size, and the next column says so. Green means a positive gap; red means negative.

Optimal size → profit. The scanner solves for the input amount that **maximises gross profit**, using the exact pool maths (BigInt, fees included). It searches within the safe-depth caps (§3.2, §3.4). The top line is that amount; the bottom line is the gross profit at that amount in the start token, followed by the return in basis points of input. When no size makes a profit, the cell reads `not profitable at any size`. When the caps leave no room for a trade, it reads `no feasible size`.

Profit (USD). The optimal-size profit converted to USD at the live token price. The small line under it, **net after gas**, subtracts the estimated gas cost: live `eth_gasPrice` × the route's gas estimate (§3.12), shown as "(~Nk gas)". **Net after gas is based on the optimal size**, not on your size. Green means it stays positive after gas.

At your size. The same full calculation, run on the exact amount in **Your trade size (start token)**. The top-line result is gross profit in the start token. The cell shows `exceeds safe depth` if your amount is beyond a leg's cap, or `–` if the size is 0.

Flags. These are labels the scanner attaches to a row:

Flag	Meaning
`candidate`	Profitable at the optimal size, at or above your min-bps, and not suspicious. Gas is not checked here; read the net-after-gas line.
`3-hop triangular`	A triangular route through three distinct pools.
`v3 multi-tick (≤N intervals)`	A V3 leg was sized across up to N tick intervals (adaptive depth, §3.9). Liquidity beyond them is not modelled.
`v3 in-interval (conservative cap)`	Tick data could not be read, so the V3 leg is limited to the current tick.
`dust depth`	A leg can absorb less than about $25. Suspicious; hidden unless *Show suspicious* is ticked.
`gap >50% — broken/ mispriced pool`	The mid gap is above 50%, which nearly always means a broken, abandoned or mispriced pool. Suspicious; hidden by default.

The profit curve: why *At your size* moves with your input

Profit depends on trade size, and it rises and then falls. A small trade barely moves the pools, so it captures most of the gap, but on a small amount. As the size grows, your own trade pushes each pool's price against you (price impact), so each extra token earns less. The **optimal size** is the peak, where one more token's gain equals the impact it causes. Past it, profit shrinks and eventually turns negative. So as you move *Your trade size* toward the optimal size, *At your size* climbs toward the optimal profit. It matches it when the two sizes are equal, and falls again if you go bigger.

Your size	At your size (illustrative numbers)
100 WPLS	+2.1
500 WPLS	+7.9
1,300 WPLS (optimal)	**+11.4** (peak)
2,000 WPLS	+8.6
3,500 WPLS	−4.0

Three details:

1. **The column updates on the next scan, not as you type.** Your size is read when a scan runs: when you click Scan, or every 15 s with Auto-refresh.

2. `exceeds safe depth`**.** V2 legs are capped at 10% of the input reserve, and V3 legs at the tick intervals read. Past a cap the scanner shows the warning instead of guessing a profit.
3. **The optimum can shift slightly between scans.** Your size sets how many V3 tick intervals are read (§3.9), and reserves change every block.

Gas costs about the same at any size, so trading near the optimum gives a route its best chance of covering gas. The dry run's measured `eth_estimateGas` and net after gas (§3.12) still decide whether a trade is worth sending.

5.2 Other displays

Display	Meaning
`not profitable at any size`	Fees are larger than the price gap at every size.
`+x TOKEN` profit	Gross profit at the optimal size, before gas.
`net after gas`	Optimal-size profit in USD minus the estimated gas cost (green = still profitable).
`v3 multi-tick (≤N intervals)`	v3 legs are sized across up to N tick intervals; liquidity beyond that is not modelled.
`v3 in-interval (conservative cap)`	Fallback when tick data could not be read: size limited to the current tick.
✅ `Would succeed` / ❌ `Would revert NoProfit`	ArbExecutor dry-run result at the latest block.
🔔 banner / notification	A new profitable, net-of-gas route ≥ min bps appeared (deduplicated per route).
`3-hop triangular`	Triangular route. Only profitable ones are listed.
Hidden: suspicious/dust	Thin or anomalous pools have been left out for your safety.

Live findings (1 Nov 2026, block ~27.94M): no profitable two-pool or triangular routes after DEX fees. The best gross gap was 0.22% (WPLS–WETH, PulseX v2 → LibertySwap V3 0.25%), which is below the combined 0.54% in fees. At about 444,000 gwei, the estimated gas is under $0.001 per route, so DEX fees, not gas, are what close the gap. The WETH bridge check showed about +0.17% to +0.30% against Coinbase before gas and slippage. Efficient markets tend to close these gaps fast, so the tool is most useful during volatility.

6. Risks and limitations

- Gas is an estimate (per-route gas units × live gas price); on the first mainnet dry run the v4.9.1 model was ~9% below measured gas; v4.9.3 recalibrated it (§3.13). Real gas still varies by tens of thousands of units per route. A tx that lands after its deadline reverts with `Expired()` and costs only the small revert gas. MEV and pending-block state are not modelled.
- v3 depth is limited to the adaptive tick depth (max 8 intervals per direction); liquidity beyond that is not modelled.

- ArbExecutor: a public-mempool tx can be front-run or the price can move after the dry run; then it reverts and gas is lost. The contract is unaudited; use small sizes first.
- Alerts run in the browser only: the tab must stay open, and the browser or OS may block notifications or sound.
- The results are a snapshot. By the time a transaction lands, reserves may have changed.
- Bridge round-trips are non-atomic and carry price risk.
- The public RPC can lag or rate-limit you.
- **This is analysis, not financial advice. Always verify on-chain before trading.**

7. Security model

By default the page is read-only: it uses only `eth_call` and HTTPS GET requests (PulseChain RPC, LibertySwap API, and, only when you click the button, the Piteas quote API). Since v4.7 there is an **opt-in** wallet connection: a transaction is sent only when you click Connect & Send *and* approve it in your wallet, and only from the executor owner. The page never sees or stores private keys; there is no tracking and no backend. All code is in one auditable HTML file.

8. Maintenance

- **Source files:** `engine.js` (engine), `ui.js` (UI logic), `body.html` (markup) and `test.js` (Node tests: `node test.js`).
- **Build:** the CSS + `body.html` + `engine.js` (without module exports) + `ui.js` are combined into one `index.html`.
- **Deploy:** commit `index.html` to `main`. GitHub Pages serves it within about a minute.
- **When fees or factories change:** re-verify on-chain and update the constants in `engine.js`.

9. Roadmap

~~first mainnet deploy~~ (done, §3.13) · first profitable live trade when a net-positive route appears · v4.10 private/bundled submission (MEV protection) · ~~v5.0 server-side Telegram alerts~~ (done, §3.14) · v5.1 keeper role + automatic execution of alerted routes · Balancer-V3 / Tide support if its liquidity grows · more venues and tokens.

10. Version history

- **v5.0.1:** Scan-log CSV (§3.14). The server scanner now appends one row per scan to `server/scan_log.csv` recording the single best *executable* route that scan — the largest mid-gap route that clears the dust floors — even when it is net-negative after gas. 19 columns (timestamp, scan, block, gas price, routes evaluated, route + pools, mid gap, optimal size/gross/net/bps, gas estimate, profit at fixed size). Git-ignored; toggled by `csv_log` / `csv_path` in `config.json` (on by default). Alert logic unchanged — this only writes a near-miss history for later analysis. Tested live: rows append once per scan.

- **v5.0:** Server scanner + Telegram alerts (§3.14). New `server/` folder: `scanner.js` runs the same engine on an always-on machine and sends a Telegram alert for every route that is net-positive after gas and not suspicious, de-duplicated per route. It only alerts and never trades. `--dry --once` lets you test without Telegram. Live dry-run scan tested (200 routes, 0 net-positive); real Telegram delivery not yet tested.
- **v4.9.4:** Documentation release. New §5.1, *Reading the results table*: every column explained (mid gap, optimal size → profit, profit USD + net after gas, at your size, flags with exact labels), the profit curve with a worked example, and the scan-timing, safe-depth and optimum-shift details. Matching in-app Docs section and README section. No scanner logic changed.
- **v4.9.3:** Gas model recalibrated on mainnet. The per-route base rises from 150k to 210k (V2 and V3 hop costs unchanged), so the V2 × 3 estimate is now 630k vs 627,914 measured. Every net-after-gas figure is slightly more conservative.
- **v4.9.2:** First mainnet deploy recorded (ArbExecutor `0xd5da…30bd`, owner verified). The live test from ZKX reached the contract but reverted `Expired()` after landing ~16.6 min late (9.73 PLS gas). Calldata deadline raised from 120 s to 600 s. Dry-run gas was 627,914 vs the 570k model estimate.
- **v4.9.1:** Gas model recalibrated from fork measurements: per-route 150k base + 140k per V2 hop + 230k per V3 hop (was flat 250k/350k, ~2.2× too low for V3 routes). Dry run now shows measured `eth_estimateGas`, PLS/USD cost and net-after-gas.
- **v4.9:** ArbExecutor keeper role (execute-only bot key; owner-only withdraw/setKeeper; profit to owner; 3 new fork tests, 14/14). Deploy kit: pinned paris build, in-app Estimate / Deploy / Set-keeper via the connected wallet; Send accepts the keeper.
- **v4.8.1:** Opt-in Piteas aggregator cross-check in Trade Companion (2 quotes, local 6/min guard, graceful "unavailable" handling); Tide venue identified on-chain as a Balancer-V3 weighted pool and documented (not scanned).
- **v4.8:** Adaptive V3 tick depth: per-pool interval count sized to the trade (1–8 cap slider, same-block reads), depth range in telemetry.
- **v4.7:** New tokens eHEX, INC (and WETH pairs) with 9 liquid pairs; opt-in MetaMask Connect & Send (chain switch, owner check, wallet-signed tx).
- **v4.6.1:** Fixed stale browser tab title (v4.3 → v4.6); white paper PDF regenerated.
- **v4.6:** ArbExecutor atomic flash-swap contract (PulseX v1/v2, 9mm, LibertySwap V3; 2–4 legs), fork-tested; Trade Companion execution panel with calldata builder and read-only dry run.
- **v4.5:** multi-tick LibertySwap V3 depth (up to 4 intervals per direction, same-block tick reads) and profitability alerts (sound + browser notification, deduplicated).
- **v4.4:** triangular 3-hop routes, gas-adjusted net profit, separate route counts in the summary, and old pages redirected to the scanner.
- **v4.3:** live Multicall3 reserves, verified fees, LibertySwap V3 and bridge quotes, and a BigInt engine.